# Elvi Zekaj

## AI/ML & Backend Engineer

📍 Tirana, Albania — **available for EU remote (CET)** & relocation ✉ elvizekaj@zedigital.tech
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### SUMMARY

Engineer who ships production systems end to end — from GPU cluster and inference infrastructure up to the client-facing product. I operate a 2-node NVIDIA DGX Blackwell cluster serving an 80B-parameter MoE model on vLLM, and build LLM-backed products for clients in Albania, Germany, Austria and the UK. 42 Heilbronn graduate, with 4,300+ GitHub contributions in the last year.

### TECHNICAL SKILLS

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|---|---|
| **Languages** | Python, TypeScript/JavaScript, C, C++, Go, Rust, SQL, Bash |
| **AI / ML** | vLLM, LLM serving & quantisation (FP8, MoE), local-first inference, RAG, vector search (FAISS, Pinecone), agent architectures, Deepgram/LiveKit speech, evaluation harnesses |
| **Infrastructure** | NVIDIA DGX / CUDA, Docker, Kubernetes, Ray, systemd, nginx, RoCE RDMA, Tailscale, VPS & bare-metal administration, CI/CD |
| **Backend** | FastAPI, Node.js, Fastify, Hono, Bun, PostgreSQL, Prisma, SQLite, REST APIs, Stripe, Twilio/SIP telephony, n8n |
| **Frontend** | React, React Native, Next.js, i18n, Tailwind |
| **Practice** | Test-driven development, mutation testing, statistical validation, observability & alerting |

### EXPERIENCE

**Founder & Engineer** 10/2025 – Present
**Z.E Digital Tech — Tirana, Albania**

- Built and operate a **2-node NVIDIA DGX GB10 (Blackwell) cluster** — 256 GB unified memory, dual ConnectX-7 200 Gbps RoCE RDMA interconnect, Ray, Docker with NVIDIA runtime, Tailscale mesh networking.
- Serve **Qwen3-Coder-Next-FP8 (80B MoE)** on vLLM at ~57 tok/s, powering internal coding, document and analysis pipelines with **zero cloud inference spend**.
- Engineered an **algorithmic trading research platform** (~23,000 lines of Python): point-in-time data layer making look-ahead bias structurally impossible, walk-forward and pre-registered hypothesis testing, **435 automated tests including 84 mutation tests**, circuit breakers and a kill-switch safety layer, deployed live against a broker API on a systemd timer fleet.
- Architected **Sofia**, a voice AI receptionist answering live hotel phone calls 24/7 in German over a SIP trunk, including a custom voice-activity detector for sub-second turn-taking (96.7% turn-end detection across 61 test scenarios).
- Delivered **18+ production projects** for clients in the UK, Austria, Germany and Albania; full lifecycle from consultation and system design through deployment and optimisation.

**IT Specialist** 05/2024 – 09/2024
**InfoSoft Systems — Tirana, Albania**

- Maintained and optimised IT infrastructure, networks and internal systems.
- Technical support, hardware/software troubleshooting, and evaluation of emerging technologies for adoption.

**IT Junior** 05/2022 – 07/2023
**One Albania — Tirana, Albania**

- Installation, configuration and maintenance of computer systems and networks for a national telecommunications operator.
- Supplier relations and hardware procurement.

## SELECTED PROJECTS

**Sofia — Voice AI Platform** Python · LiveKit · Twilio · Deepgram · Docker

Production voice agent handling live inbound phone calls 24/7 in German via SIP. Custom voice-activity detector with semantic turn-end detection — 96.7% accuracy across 61 German conversation scenarios, 0.077 ms per decision, 34 KB footprint. Deployed in concierge, telephony and sales contexts.

**Quantitative Trading Platform** Python · NumPy/pandas · SQLite · systemd

Research and execution system with a causality-enforcing data layer, deflated Sharpe and multiple-testing correction, pre-registered hypothesis gates, and a 17-switch safety layer. Runs unattended against a live broker paper account.

**ELO-AGI (NEURO)** Python · EWC · PackNet · local-first

Neuroscience-inspired cognitive framework — 38 modules across 4 reasoning tiers combining Global Workspace theory, predictive coding and dual-process reasoning. Continual learning via EWC and PackNet. Runs fully offline.

**Verto — CRM & ERP Platform** TypeScript · Docker · nginx · PostgreSQL · n8n

Multi-vertical business platform (dental, hotel, spa, clinic, law, academy, rent-a-car) on production VPS, branded per industry with custom workflows and automation.

**Mobial — eSIM SaaS** Next.js · Stripe · Prisma · React Native

Multi-tenant SaaS at mobialo.eu with payment processing, Accept-Language i18n and a companion React Native app.

**Zeo Dental** React 19 · Fastify · PostgreSQL · Gemini AI

Clinic platform with AI chatbot — 70% inquiry automation and 2× bookings. Records, scheduling and document management. Delivered in 4 weeks.

## OPEN SOURCE

**ezekaj/agielo** — cognitive AI framework, Python

**zedigitaltech/mobial** — eSIM SaaS, TypeScript

**zedigitaltech/zeo_dental** — clinic platform, TypeScript

**zedigitaltech/elo-assistant** — AI coding assistant, Python

## EDUCATION

**BSc Software Development** 09/2023 – 02/2026

**42 Heilbronn gGmbH — Heilbronn, Germany**

Peer-to-peer, project-based curriculum: C, C++, Unix systems programming, algorithms, networking, graphics.

**BSc AI Engineering** — paused

**Universiteti Metropolitan Tirana — Albania**

**Professional High School** 09/2019 – 06/2023

**"Hermann Gmeiner" — Tirana, Albania**

## LANGUAGES

**Albanian** native · **English** C2 · **German** A2 (learning)